Software Development

Course name:	Computer Science 1
Instructor(s) and contact information:	Nic Bander nic.bander@evsck12.com 812-435-8827
Textbooks:	No Textbooks
Required materials for course:	Black or Blue Ink Pen and Notepad 8GB USB Flash Drive (optional) EVSC Google Email Account EVSC Google Calendar EVSC Google Classroom EVSC Google Drive
Dual credit(s):	Computer Science I (3 credit hours each) INFM 109 – Informatics Fundamentals SDEV 120 – Computing Logic SDEV 153 – Intro to Web Development DBMS 110 – Intro to Database Management SDEV 140 – Intro to Software Development
Industry certification(s):	No certifications at this time
Student organization(s):	No certifications at this time
Internship/Practicum:	Internships are only available for 2nd semester for Computer Science 2

I. COURSE INFORMATION	
Course title(s)/description(s):	NLPS 7183 Principles of Computing provides students the opportunity to explore how computers can be used in a wide variety of settings. The course will begin by exploring trends of computing and the necessary skills to implement information systems. Topics include operating systems, database technology, cybersecurity, cloud implementations and other concepts associated with applying the principles of good information management to the organization. Students will also have the opportunity to utilize basic programming skills to develop scripts designed to solve problems. Students will learn about algorithms, logic development and flowcharting.

	NLPS 7185 Website and Database Development will provide students a basic understanding of the essential Web and Database skills and business practices that directly relate to Internet technologies used in Web site and Database design and development. Students will learn to develop Web sites using Hypertext Markup Language (HTML) and Cascading Style Sheets (CSS). Additionally students will be introduced to the basic concepts of databases including types of databases, general database environments, database design, normalization and development of tables, queries, reports, and applications. Students will be familiarized with the use of ANSI Standard Structured Query Language. Students will be introduced to data concepts such as data warehousing, data mining, and BIG Data. Students will develop a business application using database software such as Microsoft Access. NLPS 7184 Software Development introduces students to concepts and practices of programming languages and software development. Students are introduced to algorithms and development tools used to document/implement computer logic. Discusses the history of software development, the different types of programming such as real time processing, web/database applications, and different program development environments. Concepts will be applied using different programming languages, and students will develop and test working programs in an integrated system.

II. COURSE CONTENT STANDARDS / SEQUENCE

Course title(s)/description(s):	Computer Science Software Development

Semester Timeline	Topic/Unit	Standard(s)
1	Number Systems & Typing	All standards can be found in the Next Level Programs of Study Standards and Competencies
1	Computer Technology History	
3	Basic Networking & Terminal Controls	
1	College & Career Research and Resumes	
5	Programming Fundamentals	
4	Python Turtle Graphics	
3	Intro to HTML & CSS	
3	Intro to Database Management	
3	Intro to Angular Web App	
3	Intro to Flask API	
4	Web Application Development	

Major skills based projects/assignments:	Build Your Own MIT ai2 App Password Manager Python Portfolio Website ILP Web Application
Integrated literacy projects/assignments:	GitHub README Development Resume and Cover Letters
Integrated numeracy projects/assignments:	Number System Conversion Data Analytics with Python

*Instructors and/or school administrators reserve the right to modify all tasks and their associated timelines throughout the **2025-2026** school year without prior approval from students and/or parents/legal guardians.*

III. EVALUATION/GRADE PROCEDURES	
Grading scale:	A: 100 - 90 B: 89 – 80 C: 79 – 70 D: 69 – 60 F: 59 – Below
Grading period calculations:	To mimic a semester grade similar to a college experience, all grading periods in a semester will be considered continuous. The combined grading periods will account for 80% of the total semester grade.
Final exams:	Each semester's final projects are calculated as 20% of the semester grade. In this course you typically will have multiple projects which will consist of the 20%. Each project is weighted based on time and difficulty. Each semester the quantity and duration of the projects will be determined by the instructor.
Frequency of grades:	Typically at least one assignment a week will be collected.
Parent\Student gradebook access:	Grades can be checked in PowerSchool at: **https://evsc.powerschool.com/public/** or on the SICTC school site at www.sictc.com. EVSC students and parents can gain access through their partner school accounts. Non-EVSC students can receive an access code by contacting the SICTC Administration Office.
Examples of assessments:	Presentations Quizzes/Exams Programs and Programming Problem Sets Projects Class productivity *(This is a subjective grade issued by the instructor to reward those that perform in a professional and responsible manner within the class, i.e. following classroom guidelines, arriving to class on time, time management, after hours messaging etc.)* Final Exam/Project *(This will count for 20% of the final semester grade and cannot be exempted for any reason.)*

Academic dishonesty policy:	If a student(s) is found using another person's or AI work, idea, or computer file, in part or in whole, the following actions will be taken: ● First Offense – A call/email home to the parent/guardian by the instructor and a zero '0' will be issued on the assignment. ● Second/Future Offense – A call/email home to the parent/guardian by the instructor, a zero '0' will be issued on the assignment and the student will be sent to the office on a disciplinary referral. If a student(s) is found aiding in the plagiarism process of another student(s), the following actions will be taken: ● First Offense – A call/email home to the parent/guardian by the instructor and a 50% reduction on the assignment. ● Second/Future Offense – A call/email home to the parent/guardian by the instructor, a zero '0' will be issued on the assignment and the student will be sent to the office on a disciplinary referral.
Homework/Project deadline policy:	Project deadlines require students to submit all work by a specific date, time, and method arranged by the instructor. A 30% deduction will be applied to the work if it is late unless one of the following scenarios applies: ● Another arrangement has been made and agreed to by students and teacher via email before the due date. ● The assignment deadline falls within ten days of the end of a grading period/semester. If a project is not turned in before the project's specified due date and time, and specific method of submission as outlined by the assignment criteria, no points will be given for the project. **Electronic loss of information is not a valid reason to miss a deadline. Backup information routinely to another storage device (Flash Drive, GitHub, Google Drive).**

Classroom Expectations	1. Respect and assist one another 2. Willing to step out of comfort zone 3. Arrive to class on time or notify the teacher if you will be late or absent 4. Be an active listener 5. Ask questions during lecture and project time and after class 6. Complete all assignments to the best of YOUR ability and if you need assistance to reach out or research the issue. Again, if you receive help be sure to give credit where credit is due. 7. Use electronic tools in appropriate behavior Based on the above guidelines, any violation against them may result in classroom privileges being suspended for a defined timeframe, dean referral, or removal from the technical program. All students must adhere to the rules and regulations set by the Evansville Vanderburgh School Corporation and the Southern Indiana Career & Technical Center. These can be found at: **https://district.evscschools.com/cms/One.aspx?portalId=74772&pageId=5508269**

IV. ATTENDANCE	
SICTC Attendance Policy:	Students are expected to attend school regularly to maximize the benefits of the instructional program. All students, including non-EVSC students, are expected to attend classes when the schools associated with the Evansville Vanderburgh School Corporation are in session. **Car rider/driver classroom start time: 7:30 AM. If the student arrives after 7:30 AM, they will be counted as late to class.**

V. ELECTRONIC DEVICES	
Personal electronic device policy:	Please refer to the EVSC Wireless Communication Device policy for further details. This can be found at: https://go.boarddocs.com/in/evsc/Board.nsf/goto?open&id=D5VRP46D6655
School-issued electronic classroom devices:	School-issued electronic devices, such as computers and tablets, are to be used only for course-related purposes within the Southern Indiana Career & Technical Center. Unauthorized use,

	such as personal email, online chatting, online gaming, installing/uninstalling software, writing or distributing viruses, downloading/playing movies and music, playing VR games, or interacting on social media (e.g., Facebook, Twitter/X, TikTok, Instagram, Snapchat, Reddit), will result in the temporary loss of computer privileges, suspension from class, counselor referrals, or removal from the program. **Any student found intentionally damaging or destroying a computer will be sent to the SICTC Office for disciplinary action and possible removal from the program.**

VI. SAFETY	
General lab safety:	No running, practical jokes, or horseplay are allowed in the lab areas at any time.
REQUIRED Personal Protective Equipment (PPE):	Students are responsible for supplying, storing, and maintaining personal protective equipment (PPE) for lab areas at the SICTC. **These items are required to ensure the safety of both the student and their peers.** Students in the lab environment must adhere to the following PPE expectations: ● Clear, plastic, anti-fog safety glasses; ANSI Z87.1 certified. **Safety glasses must be worn at all times in the lab area.** No tinted or mirrored safety glasses are allowed. ● Clothing appropriate and consistent with industry standards, while adhering to school dress code. Example: No shorts are allowed in the lab area. ● Leather work boots that cover at least the ankle, preferably with a steel toe (not required). **No open-toed shoes or tennis shoes.** ● No baggy clothing or jewelry (chains, rings, etc.). ● Hard hat to be worn at all times in the lab area. *NOTE: Additional PPE may be required for specific activities and will be discussed with the student or provided by the SICTC before starting the activity.*
Personal electronic devices:	No personal electronic devices are allowed in the lab area unless specifically permitted by the instructor. Please refer to the EVSC Wireless Communication Device policy for further details. This can be found at: https://go.boarddocs.com/in/evsc/Board.nsf/goto?open&id=D5VRP46D6655

Food and beverages:	Students may have food and beverages only in designated lab areas and at designated times. Food is only allowed during break time or when the teacher states it is acceptable. Drinks are allowed in class, but a **lid/cap must** be on the container when you are not drinking. Remember, if you spill a drink on a device and ruin it, you are responsible to replace the damaged components or device. If a mess was made, please clean it quickly. If messes are left after class, the class as a whole gets two strikes before all food and drinks are no longer prohibited in class.
Safety procedures and cleanliness:	Students must follow all written and verbal safety procedures and keep the lab clean at all times. Failure to comply may result in losing lab privileges, receiving a referral to the SICTC office, or being removed from the program. All students must adhere to the rules and regulations set by the Evansville Vanderburgh School Corporation and the Southern Indiana Career & Technical Center. These can be found at: **https://district.evscschools.com/cms/One.aspx?portalId=74772&pageId=5508269**

VII. SPECIFIC CLASSROOM INFORMATION	
Entering Classroom:	1. Come in, obtain your laptop, and go directly to your assigned seat. 2. Once in your seat, be ready to begin class when the bell rings. This includes but is not limited to: having your computer on, having any homework completed, picking up any handouts, and beginning on any work from the previous day, bell ringers, or as directed. 3. Please check the board for any bell ringers or pertinent information.
Ending & Exiting Class:	Five minutes before class ends, we will start cleaning up your workspace. This includes logging off your laptop, returning the laptop, and cleaning any mess. All the laptops and equipment must be turned off and organized on the lab desks. Once all equipment is put up, you must also make sure your work area is free of trash or material that was not there before you came to class. Chairs under the table, and tabletops clean. If your laptop was left out without the teacher's permission, you will be responsible to put everyone's laptop up the next school day.

Classroom Computer Usage:	The computers within the classroom are to be utilized for purposes relating to course subjects taught within the Southern Indiana Career & Technical Center. Unauthorized use of the computer for other topics or operations will result in the temporary loss of the computer privileges, suspension from class, counselor referrals, or removal from the program. Examples of unauthorized use: online chatting/discussion board (other than those directed by teacher), gaming, unauthorized installing or uninstalling software, writing or distributing virus applications or other malware, downloading/playing movies and music and interacting on social media websites during class such as social media, gaming, etc. ANY STUDENT FOUND INTENTIONALLY DAMAGING OR DESTROYING A COMPUTER WITHIN THE CLASSROOM WILL BE SENT TO THE PRINCIPAL FOR DISCIPLINARY ACTIONS AND POSSIBLE REMOVAL FROM THE PROGRAM.
Artificial intelligence (AI) usage policy:	In this course, you may use generative artificial intelligence (AI) tools, such as ChatGPT, MidJourney, Canva, Fireflies, etc., for **assistance only**. These tools can be helpful when you're stuck or need a starting point, but they should **not** replace your own critical thinking and effort. **Any use of AI tools, even for planning or brainstorming, must be clearly acknowledged and cited.** You are responsible for reviewing information on how to properly cite AI tools in the style most commonly used in the computer science industry (e.g., APA style). **Here's what you need to do when using AI for assistance:** **Cite the AI tool(s) used:** Clearly state which AI tool(s) you used (e.g., "Assisted by ChatGPT," "Image generated with MidJourney"). ● **Explain the specific assistance received:** Describe exactly how the AI tool helped you. For example, "ChatGPT was used to brainstorm initial ideas for the algorithm," or "MidJourney generated an image for the user interface, which was then modified." ● **Acknowledge the limitations:** Remember that AI tools can make errors, generate false information, or potentially infringe on intellectual property rights. Always **"trust but verify"** any output from an AI tool, and ensure the final work is your own and accurate. **Important Information Regarding US Copyright and Trademark Laws:** ● **Copyright and AI-Generated Content:** Under current U.S. copyright law, copyright protection generally requires **human**

	authorship. This means that purely AI-generated content, without significant human creative input and control over its expressive elements, is typically **not eligible for copyright protection**. If you use AI as an assistive tool, your original human contributions to the work may still be copyrightable. However, if the AI determines the expressive elements of the output, that portion of the work may not be copyrightable. Be mindful that using copyrighted material to train AI models is a developing area of law, and there are ongoing legal discussions about whether such use constitutes copyright infringement. ● **Trademark and AI-Generated Content:** While copyright often focuses on human authorship, trademark law is different. A **trademark** protects brand names, logos, slogans, and other identifiers used in commerce to distinguish goods and services. If you use an AI tool to generate a logo or brand element, and you use that element in commerce to identify your goods or services, it *may* be eligible for trademark protection, regardless of its AI origin. However, you must ensure that any AI-generated content does not infringe on existing trademarks of other entities. AI tools may unintentionally generate content that is confusingly similar to existing trademarks. ● **Your Responsibility:** It is your responsibility to understand and adhere to U.S. copyright and trademark laws. **Do not use AI tools to generate content that may infringe on the intellectual property rights of others.** This includes, but is not limited to, generating images, text, or code that closely resembles existing copyrighted or trademarked works without proper authorization. **Please be aware:** **Grade Reduction for AI Use:** While using AI for assistance is permitted and encouraged when you're stuck, there will be a **grade reduction** for any work that incorporates AI-generated content. This reduction will be less severe if you properly comment on and cite your use of AI, as outlined above. ● **Plagiarism and Intellectual Property Infringement:** Any **uncited or unacknowledged use** of AI tools, or any use that results in the submission of work that infringes on copyright or trademark laws, will be considered **plagiarism and a violation of academic integrity**. This is a serious academic offense and will be subject to the academic dishonesty policy outlined in this syllabus.

VIII. IMPORTANT DATES	
EVSC/SICTC School Begins (students with last names K-Z)	August 4-5, 2025
EVSC/SICTC School Begins (students with last names A-J)	August 6-7, 2025
EVSC/SICTC School Begins (all students)	August 8, 2025
EVSC/SICTC No School – Labor Day	September 1, 2025
EVSC/SICTC No School - EVSC Staff Development Day	September 24, 2025
EVSC/SICTC End of First 9-week Grading Period	October 3, 2025
EVSC/SICTC Fall Break	October 13-17, 2025
SICTC Open House (Weeknight)	November 5, 2025
SICTC College and Career Fair	November 14, 2025
SICTC Open House (Saturday)	November 15, 2025
EVSC/SICTC Thanksgiving Break	November 26-28, 2025
SICTC Finals Week	December 8-12, 2025
EVSC/SICTC End of Second 9-week Grading Period/First Semester	December 19, 2025
EVSC/SICTC Winter Break	December 22, 2025 -January 2, 2026
	January 5, 2026
EVSC/SICTC No School - EVSC Staff Development Day	January 6, 2026
EVSC/SICTC First Day of Second Semester	January 19, 2026
EVSC/SICTC No School – MLK Jr. Day	February 16, 2026
EVSC/SICTC No School – President's Day	February 27, 2026
EVSC/SICTC No School - EVSC Potential Makeup Day	March 11, 2026
EVSC/SICTC No School - EVSC Staff Development Day	March 13, 2026
EVSC/SICTC End of Third 9-week Grading Period	March 23-27, 2026
EVSC/SICTC Spring Break	April 10, 2026
EVSC/SICTC No School (Potential Makeup Day)	TBD
SICTC Class Day(s)	May 1, 2026
EVSC/SICTC No School (Potential Makeup Day)	May 4-8, 2026
SICTC Finals Week	May 22, 2026
EVSC/SICTC End of Fourth 9-week Grading Period/Second Semester	

Software Development

Please sign and return the following documents to the instructor and retain the above information for future reference.

Statement of Confirmation:

By signing below, I confirm that I have read and understand the course syllabus information related to Software Development at the Southern Indiana Career & Technical Center. This agreement will serve as a common means of class operation until the end of the 2025-2026 school year.

Student Equipment Usage Agreement:

By signing below, I acknowledge that I understand I am responsible for the care of any classroom equipment assigned to me. This responsibility includes, but is not limited to, the replacement cost of lost or stolen items and repair costs associated with misuse or damage. Failure to uphold this responsibility may result in suspension of equipment privileges or disciplinary action by the Southern Indiana Career & Technical Center administration.

Student Printed Name: _____

Student Signature: _____ Date: ____/____/____

Student Phone Number: _____

Software Development

Please have your parent/guardian(s) sign the following and return it to the instructor along with your copy of the ‘Statement of Confirmation’ and the ‘Student Equipment Usage Agreement’.

Statement of Confirmation:

By signing below, I confirm that I have read and understand the course syllabus information related to Software Development at the Southern Indiana Career & Technical Center. This agreement will serve as a common means of class operation until the end of the 2025-2026 school year.

Student Equipment Usage Agreement:

By signing below, I confirm that I have read and understand the ‘Student Equipment Usage Agreement’ and authorize my student to participate in the equipment loan program. I accept all costs associated with the repair or replacement of any items of similar brand, model, and features.

Parent/Guardian Printed Name: _____

Parent/Guardian Signature: _____ Date: ____/____/____

Parent/Guardian Phone Number: _____